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Aftertreatment Sensor Interchangeability Matrix

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Products Affected

- B6.7 CM2350 B121B
- L9 CM2350 L116B
- X15 CM2350 X114B
- X15 CM2350 X116B

Customer Communication

- This document describes the interchangeability of the aftertreatment exhaust gas temperature sensor, aftertreatment outlet NOx sensor, and aftertreatment particulate sensor on the EPA 2017 aftertreatment system.
- The below tables can be used to determine the interchangeability of the sensors used on the aftertreatment system.
- Part number differences are based off of wire lead lengths **only**. Therefore, a sensor with a longer lead length can be used in place of a sensor with a shorter lead length as long as the extra wiring is properly coiled and secured wires do **not** come into contact with hot components and/or areas of wear that can result in damage. See Figure 1 below.



Figure 1, Proper Wire Coiling.

Aftertreatment Exhaust Gas Temperature Sensor:

- The aftertreatment exhaust gas temperature sensor is a one-piece unit made up of four sensor probes.
- The aftertreatment temperature sensor module and aftertreatment temperature sensor probes are **not** serviceable separately and **must** be replaced as an assembly.
- Aftertreatment exhaust gas temperature sensor modules are **not** voltage specific and will work on 12v and 24v applications.
- Aftertreatment exhaust gas temperature sensor module probes have unique thread sizes to confirm installation in correct temperature port.
- Horizontal and vertical aftertreatment exhaust gas temperature sensor modules are **not** able to be interchanged as the probe thread sizes are different.
- See Figure 1 below for aftertreatment exhaust gas temperature sensor probe locations.

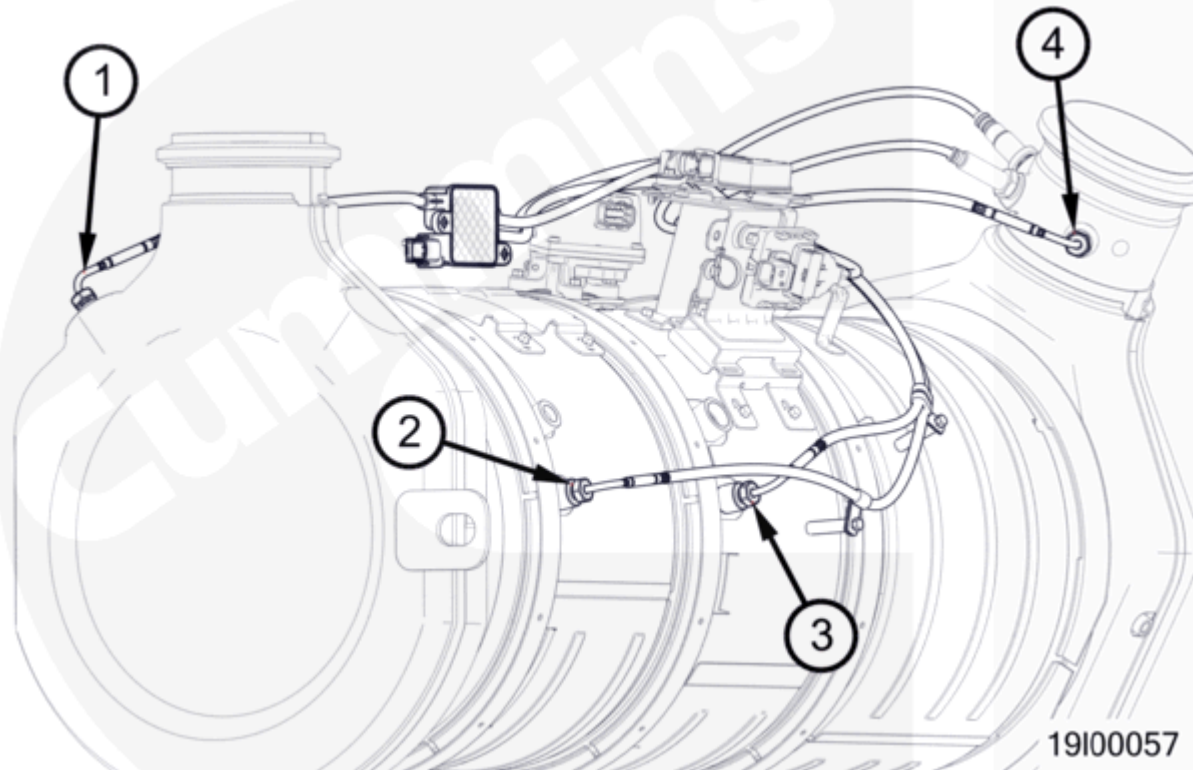


Figure 2, Aftertreatment Exhaust Gas Temperature Sensor Probe Locations.

1. T1: Intake of aftertreatment diesel oxidation catalyst (DOC)
2. T2: Intake of aftertreatment diesel particulate filter (DPF) (DOC outlet)
3. T3: Outlet of aftertreatment DPF
4. T4: Outlet of aftertreatment selective catalytic reduction (SCR) catalyst

Table 1, Aftertreatment Exhaust Gas Temperature Sensor for Horizontal Applications

Part Number	Voltage	T1 Lead Length	T2 Lead Length	T3 Lead Length	T4 Lead Length
4384561	12v/24v	340 mm	1100 mm	900 mm	750 mm
4384562	12v/24v	320 mm	1100 mm	900 mm	990 mm
4384563	12v/24v	340 mm	1100 mm	900 mm	1100 mm
4384564	12v/24v	340 mm	1100 mm	900 mm	1190 mm
4384565	12v/24v	460 mm	1100 mm	900 mm	940 mm
4384566	12v/24v	580 mm	1100 mm	900 mm	1140 mm
4384567	12v/24v	680 mm	1100 mm	900 mm	990 mm
4384568	12v/24v	690 mm	1100 mm	900 mm	1110 mm
4384569	12v/24v	670 mm	1100 mm	900 mm	1220 mm
4384570	12v/24v	630 mm	1100 mm	900 mm	690 mm
4384571	12v/24v	680 mm	1100 mm	900 mm	910 mm
4384572	12v/24v	820 mm	1100 mm	900 mm	1030 mm
4384573	12v/24v	330 mm	555 mm	355 mm	840 mm
4384574	12v/24v	435 mm	555 mm	355 mm	890 mm

Table 1, Aftertreatment Exhaust Gas Temperature Sensor for Horizontal Applications

Part Number	Voltage	T1 Lead Length	T2 Lead Length	T3 Lead Length	T4 Lead Length
4384575	12v/24v	950 mm	1100 mm	900 mm	1300 mm
4384576	12v/24v	1200 mm	1100 mm	900 mm	1300 mm
4384579	12v/24v	1700 mm	1300 mm	1100 mm	1700 mm
4384580	12v/24v	1700 mm	1700 mm	1700 mm	1700 mm

Table 2, Aftertreatment Exhaust Gas Temperature Sensor for Vertical Applications

Part Number	Voltage	T1 Lead Length	T2 Lead Length	T3 Lead Length	T4 Lead Length
4384577	12v/24v	1500 mm	650 mm	450 mm	1100 mm
4384578	12v/24v	1500 mm	700 mm	450 mm	850 mm

Aftertreatment System-Out NOx Sensor:

- NOx sensors have unique part numbers for 12v and 24v applications.

Table 3, Aftertreatment Outlet NOx Sensor for 12v Applications

Part Number	Voltage	Lead Length
4326869	12v	415 mm
4326872	12v	615 mm
4326874	12v	915 mm

Table 4, Aftertreatment Outlet NOx Sensor for 24v Applications

Part Number	Voltage	Lead Length
4326861	24v	415 mm
4326864	24v	615 mm
4326868	24v	915 mm

Particulate Matter Sensor:

- Aftertreatment particulate sensors have unique part numbers for 12v and 24v applications.

Table 5, Aftertreatment Particulate Sensor for 12v Applications

Part Number	Voltage	Lead Length
4384375	12v	600 mm
4384582	12v	800 mm

Table 5, Aftertreatment Particulate Sensor for 12v Applications		
Part Number	Voltage	Lead Length
4384376	12v	1000 mm
4384377	12v	1200 mm
Table 6, Aftertreatment Particulate Sensor for 24v Applications		
Part Number	Voltage	Lead Length
4384378	24v	600mm
4384821	24v	800mm
4384379	24v	1000mm
4384380	24v	1200mm

Document History

Date	Details
2017-4-10	Module Created